Mathematics: Fun tricks for everybody

Mathematics from high school to university

Derivation of a formula, Problem 2

Hania Uscka-Wehlou, Ph.D. (2009, Uppsala University: Mathematics)

University teacher in mathematics, Sweden

Man in The Middle AB

Problem 2: Derive the following formula for $n \in \mathbb{N}^+$,

and show a geometrical illustration of this formula:

$$\sum_{k=1}^{n}(2k-1) = 1+3+5+\cdots+(2n-1) = n^2.$$

Problem 2: Derive the following formula for $n \in \mathbb{N}^+$,

and show a geometrical illustration of this formula:

$$\sum_{k=1}^{n}(2k-1) = 1+3+5+\cdots+(2n-1) = n^2.$$

This is how the formula looks for some values of n:

$n = 1:\quad 1 = 1^2 = 1$

$n = 2:\quad 1+3 = 2^2 = 4$

$n = 3:\quad 1+3+5 = 3^2 = 9$

$n = 4:\quad 1+3+5+7 = 4^2 = 16$

$n = 5:\quad 1+3+5+7+9 = 5^2 = 25$

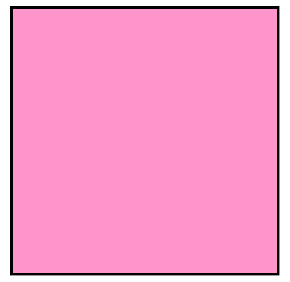
$n = 6:\quad 1+3+5+7+9+11 = 6^2 = 36$

$n = 7:\quad 1+3+5+7+9+11+13 = 7^2 = 49$

$n = 6$

$$1 + 3 + 5 + 7 + 9 + 11 = 6^2$$

$n = 6$



$$1 + 3 + 5 + 7 + 9 + 11 = 6^2$$

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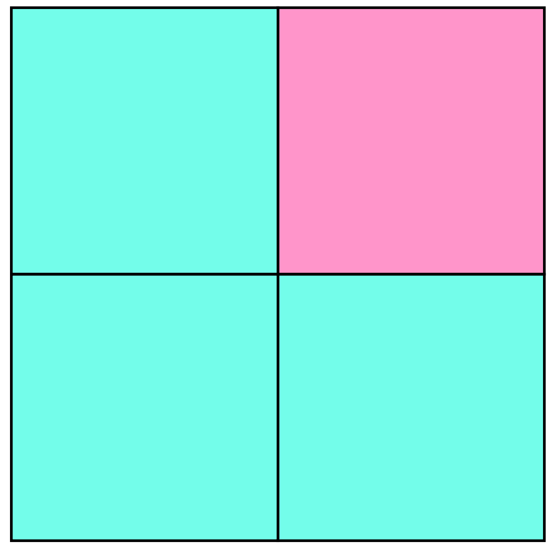
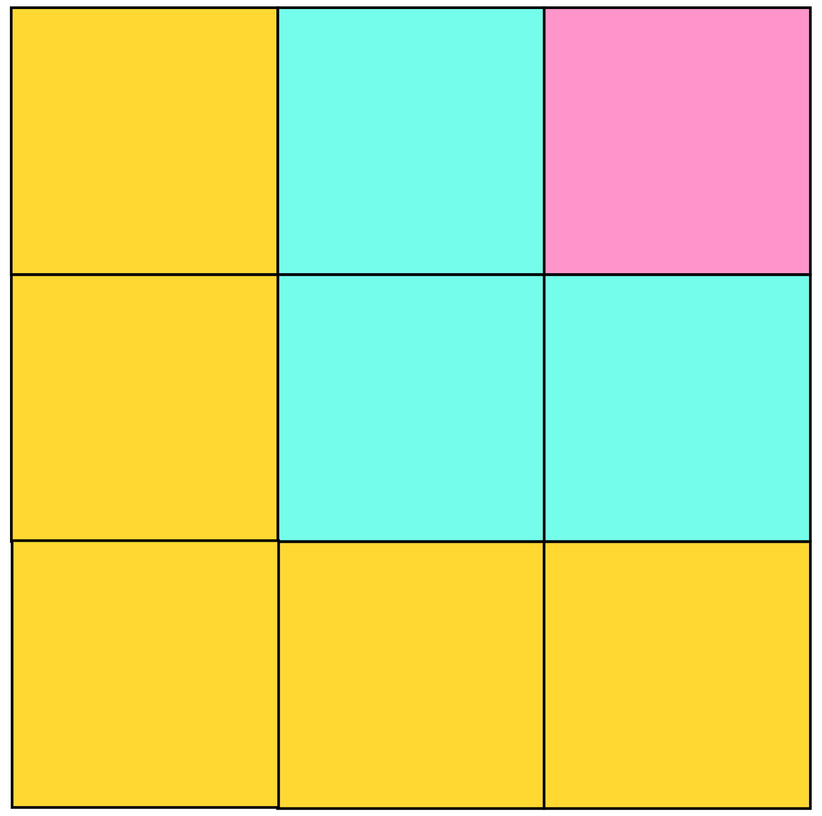


$$\boxed{1} + \boxed{3} + 5 + 7 + 9 + 11 = 6^2$$

$n = 6$



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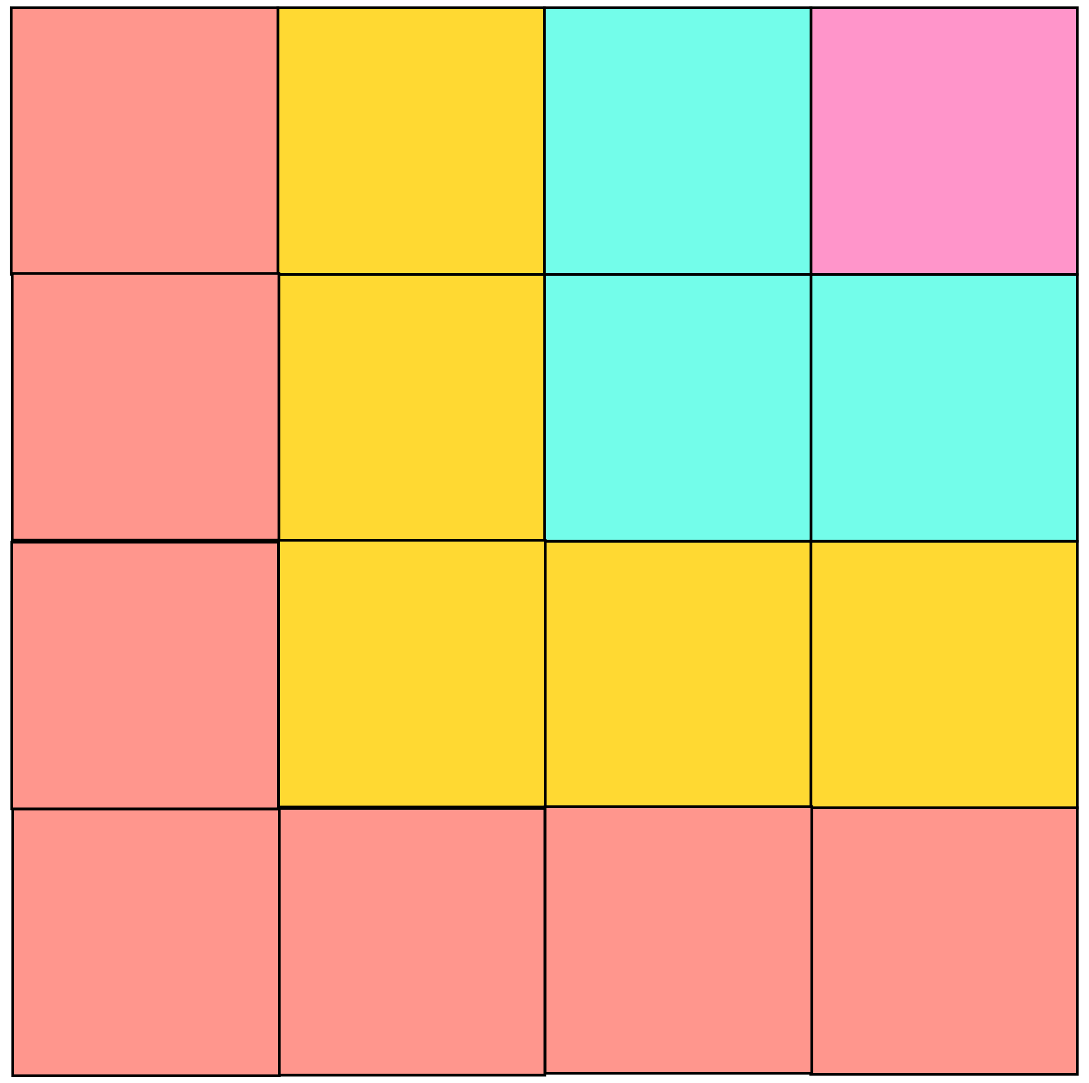

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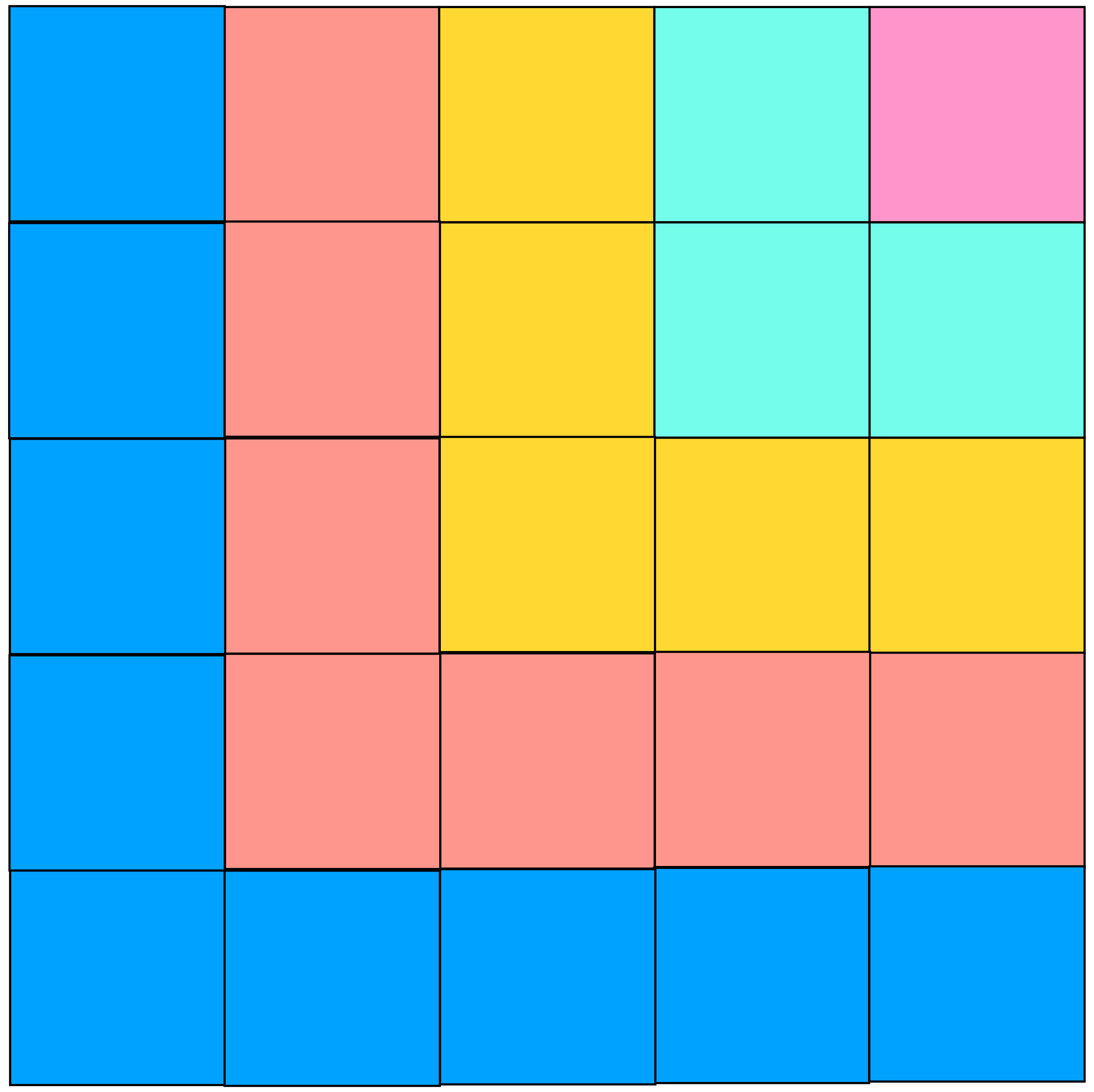

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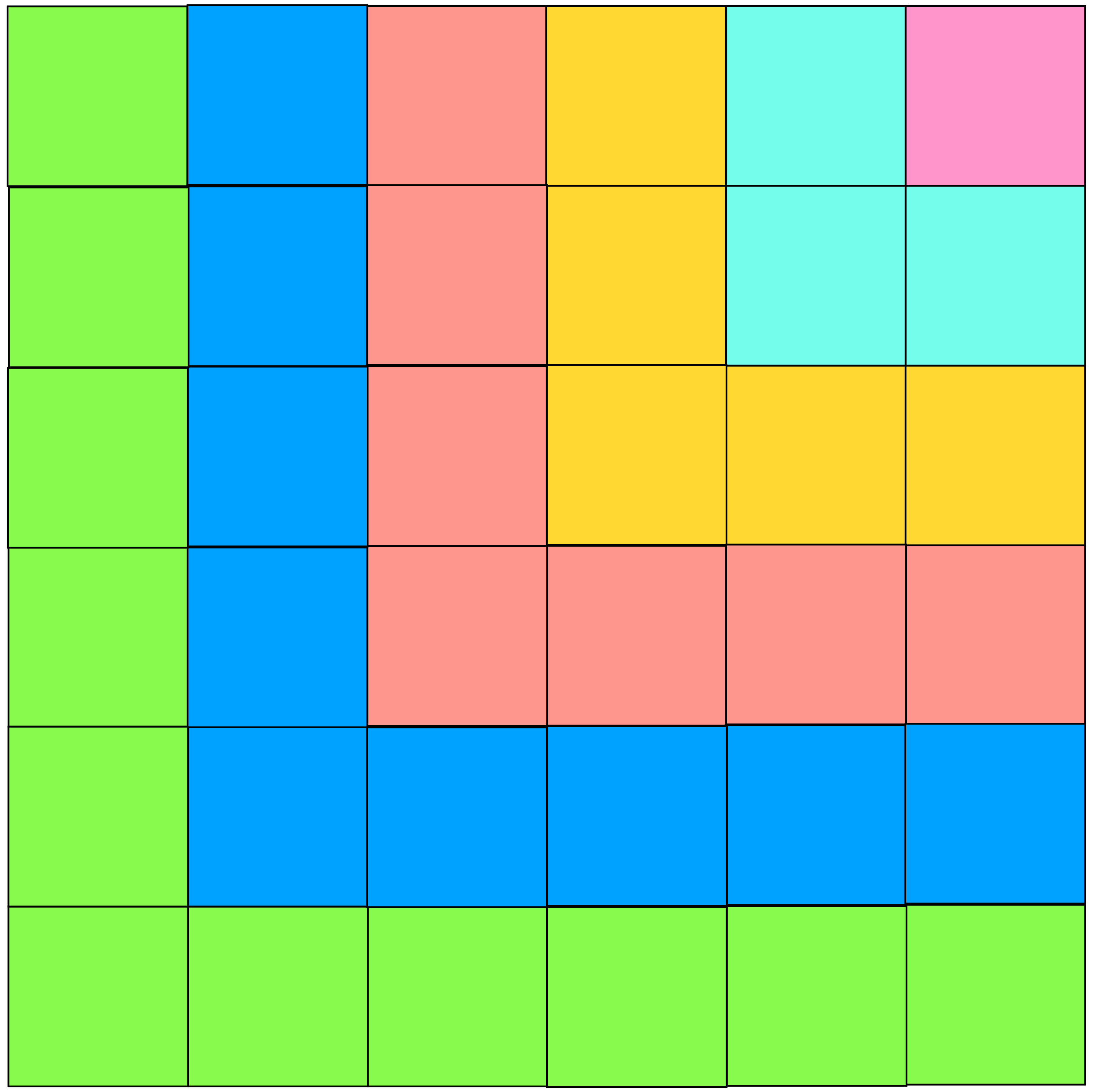

$$1 + 3 + 5 + 7 + 9 + 11 = 6^2$$

Sequence a_n and its *partial sums* S_n

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$$a_1, \quad a_2, \quad a_3, \quad a_4, \quad a_5, \quad a_6, \quad a_7, \quad \ldots$$

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$$a_n = n: \quad 1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9,\ \ldots$$

Lecture 2

Sequence a_n and its *partial sums* S_n

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$$a_n = n: \quad 1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9,\ \ldots$$

Lecture 2

$$a_n = 2n - 1: \quad 1,\ 3,\ 5,\ 7,\ 9,\ 11,\ 13,\ 15,\ 17,\ \ldots$$

Lecture 3

Sequence a_n and its *partial sums* S_n

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$a_n = n^2$: $1,\ 4,\ 9,\ 16,\ 25,\ 36,\ 49,\ 64,\ 81,\ \ldots$ Lecture 4

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Lecture 4

$$a_n = \frac{1}{n(n+1)}:\quad \frac{1}{1\cdot 2},\ \frac{1}{2\cdot 3},\ \frac{1}{3\cdot 4},\ \frac{1}{4\cdot 5},\ \ldots$$

Lecture 6

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$$a_n = \frac{1}{2^n}: \quad \frac{1}{2},\ \frac{1}{2^2},\ \frac{1}{2^3},\ \frac{1}{2^4},\ \frac{1}{2^5},\ \frac{1}{2^6},\ \frac{1}{2^7},\ \frac{1}{2^8},\ \dots$$

Lecture 7

Sequence a_n and its *partial sums* S_n

$$a_1, \quad a_2, \quad a_3, \quad a_4, \quad a_5, \quad a_6, \quad a_7, \quad \ldots$$

$$S_1 = a_1$$

Sequence a_n and its *partial sums* S_n

$$a_1,\ a_2,\ a_3,\ a_4,\ a_5,\ a_6,\ a_7,\ \ldots$$

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

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$$\cdots$$

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Arithmetic progressions

$$a_n = n: \quad 1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9,\ \ldots$$

Lecture 2

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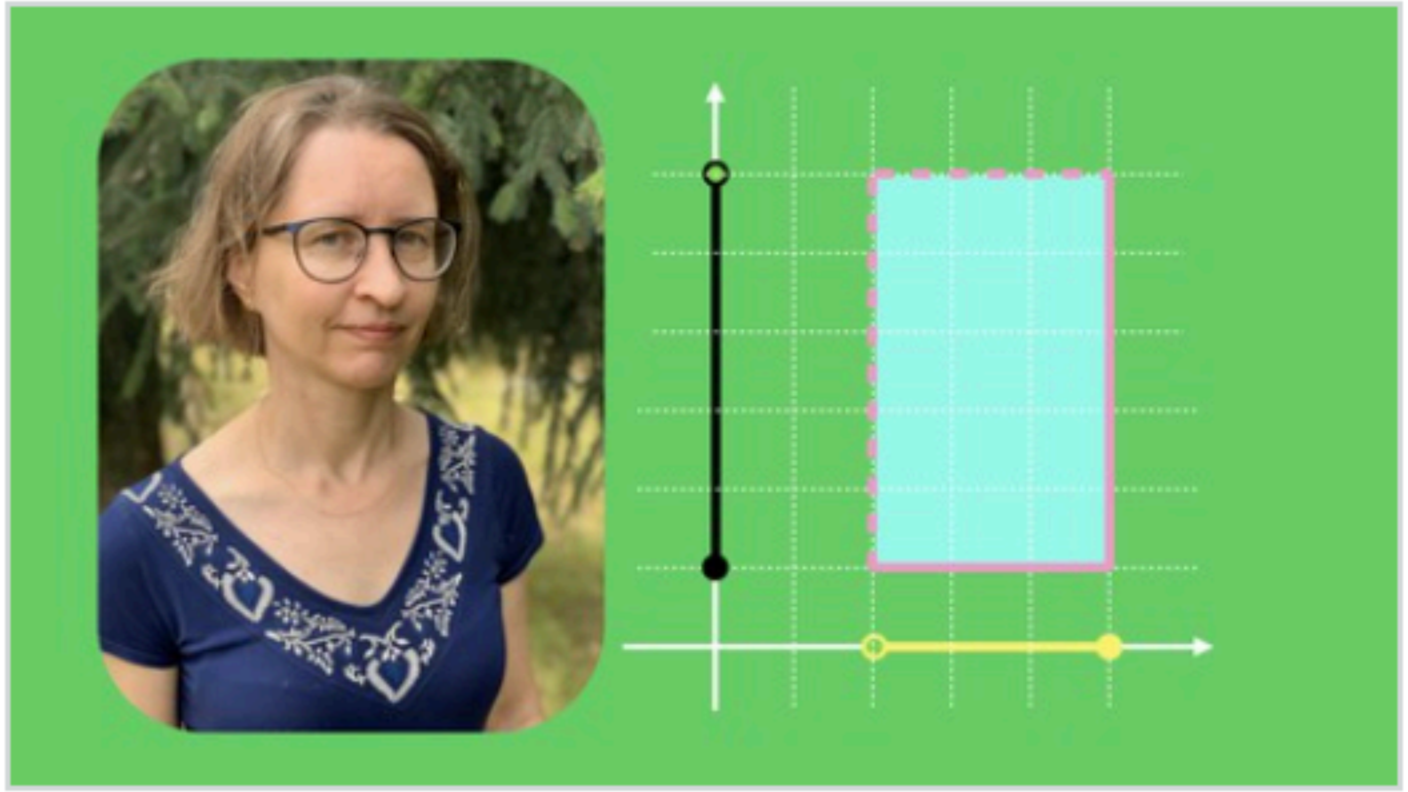
Lecture 7

Video 232



Precalculus 1: Basic notions

Mathematics from high school to university

Hania Uscka-Wehlou, Martin Wehlou

4.8 ★★★★★ (130)

52 total hours · 238 lectures · All Levels

Bestseller

Derive the formula for the partial sum S_n for the sequence $a_n = a + (n-1)d$.

$$a_1 = a,\ \ a_2 = a + d,\ \ a_3 = a + 2d,\ \ a_4 = a + 3d,\ \ a_5 = a + 4d,\ \ \ldots,\ \ a_n = a + (n-1)d$$

$$\begin{array}{rcrcrcrcr} S_n & = & a & + & a+d & +\cdots+ & a+(n-2)d & + & a+(n-1)d \\ S_n & = & a+(n-1)d & + & a+(n-2)d & +\cdots+ & a+d & + & a \\ \hline 2S_n & = & 2a+(n-1)d & + & 2a+(n-1)d & +\cdots+ & 2a+(n-1)d & + & 2a+(n-1)d \end{array}$$

$$2S_n = 2na + n(n-1)d \quad \Rightarrow \quad \boxed{\sum_{k=1}^{n} a_k = na + d \cdot \frac{(n-1)n}{2}}.$$

Video 232

Problem 6

$$1 + 3 + 5 + 7 + \dots + (2n-1)$$

$+2 \quad +2$

From the formula

① $1 = 2\cdot 1 - 1$

② $3 = 2\cdot 2 - 1$

⋮

ⓝ $2\cdot n - 1$

$a = 1 \qquad d = 2$

$$S_n = n + \cancel{2}\,\frac{(n-1)n}{\cancel{2}} =$$

d

$$= n + n(n-1) =$$

$$= n(\cancel{1} + (n-\cancel{1})) =$$

$$= n^2$$

Method 2

We know that $\boxed{1+2+\dots+n = \frac{n(n+1)}{2}}$

$$1 + 3 + 5 + \dots + (2n-1) =$$

Lecture 4

Lecture 4 Problem 3

Derive the following formula for $n \in \mathbb{N}^+$:

$$\boxed{S_n^{(2)} = \sum_{k=1}^{n} k^2 = 1+4+9+\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}}$$

SOLUTION

I will show you a really cool tricks, which will help you derive similar formulas for natural $j \geq 2$

$$\boxed{S_n^{(j)} = \sum_{k=1}^{n} k^j = 1+2^j+3^j+\ldots+n^j}$$

provided that we know all the formulas $S_n^{(i)}$ for $i = 1, 2, \ldots, j-1$.

By now we know $S_n^{(1)}$:

$$(1) \quad \boxed{S_n^{(1)} = \sum_{k=1}^{n} k = 1+2+\ldots+n = \frac{n(n+1)}{2}}$$

• • •

$$= (n+1) \cdot \frac{2n^2+n}{2} = \boxed{\frac{n(n+1)(2n+1)}{2}}$$

Division of both sides by 3 gives:

$$\boxed{S_n^{(2)} = \frac{n(n+1)(2n+1)}{6}}$$

We are done here.

You can prove in a similar way (but using the formulas for $S_n^{(1)}$ and $S_n^{(2)}$ together with the formula

with $b=1$ and varying a $\leftarrow$ $(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$)

that:

$$S_n^{(3)} = \sum_{k=1}^{n} k^3 = 1^3+2^3+\ldots+n^3 = \left(\frac{n(n+1)}{2}\right)^2.$$